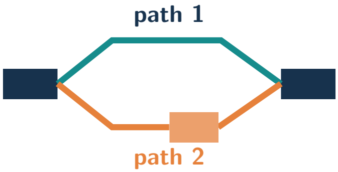
Differential splicing from equivalence-class counts

Local splice-path effects propagated through isoforms and primer-aware observations

The target is a local splice choice with biological replication



splice block between constitutive anchors

Two questions

Cell type, does path usage differ across cell types while condition is unrestricted?

Condition, does path usage differ across conditions within one cell type?

- One hypothesis per nonredundant local path partition
- Mouse is the biological replication unit
- Cells contribute to mouse $\times$ condition $\times$ cell-type pseudobulks, never independent replicates

Forward model, splice-path effects → isoforms → ECs → counts

Local path effects

$\boldsymbol{\delta}_{bmc} \in \mathbb{R}^{d_b}$
$d_b = S_b - 1$

perturb pooled isoform baseline

→

Isoform composition

$\boldsymbol{\eta}_{gi,-} \in \mathbb{R}^{T_g-1}$
$\boldsymbol{\theta}_{gi} = \mathrm{softmax}\{(\boldsymbol{\eta}'_{gi,-}, 0)'\}$

shared across primers

→

EC probabilities

$A_{gp} \in \mathbb{R}_+^{K_{gp} \times T_g}$
$\mathbf{q}_{gip} = \dfrac{A_{gp}\boldsymbol{\theta}_{gi}}{\mathbf{1}' A_{gp}\boldsymbol{\theta}_{gi}}$

primer-specific maps

→

Observed counts

$\mathbf{c}_{gip} \in \mathbb{N}_0^{K_{gp}}$
$\mathbf{c}_{gip} \sim \mathrm{Multinomial}(n_{gip}, \mathbf{q}_{gip})$

paired oligo(dT) and random-hexamer vectors

Ambiguous ECs remain ambiguous, primer design enters through A_{gp}, and the same pooled-baseline path perturbation generates both primer observations.

Direct local-path inference uses biological replicates

One local fit per mouse and level

Within-path isoform ratios and nuisance isoforms are fixed at a label-independent pooled baseline. The $d_b = S_b - 1$ path coordinates are estimated from both primer-specific EC vectors with two pseudo-observations per path.

For cell types, mouse m contributes the paired path-ILR difference $\widehat{\mathbf{d}}_{bm} = \widehat{\mathbf{y}}_{bm2} - \widehat{\mathbf{y}}_{bm1} \in \mathbb{R}^{d_b}$.

- Paired t test for two paths, Hotelling test for more paths
- Robust variance moderation for scalar tests
- Multivariate mouse-level regression for condition within cell type

Estimate
mouse-level path ILRs

aggregate
within mouse

Test
paired or independent mice

32 design-matched nulls
empirical tail calibration

Calibration matters

Cell-type labels are sign-flipped within mouse; condition labels are permuted across mice. The complete scoring pipeline is rerun for every null family.

Two brain datasets, one subject-level analysis design

Mouse

GSE189033

Biology, 5xFAD amyloidosis crossed with Csf1r ΔFIRE microglial deficiency, including PBS and microglia-transplant arms

Samples, 48 mice, six groups, balanced 4 female and 4 male per group, age 5–6 months

Assay, SPLiT-seq single-nucleus RNA-seq, paired oligo(dT) and random-hexamer observations

Analysis strata, 936 mouse × condition × cell-type pseudobulks before feature-level support filtering

Human

GSE233208

Biology, cortical postmortem tissue spanning control, sporadic Alzheimer disease, Down syndrome, and Down syndrome with AD

Scale, 396,277 annotated nuclei represented by 40 paired sequencing runs in five batches

Assay, Parse Biosciences Evercode WT v1 single-nucleus RNA-seq

Role, independent cross-species computational sensitivity for the former global EC GLMM

Sources, Spangenberg et al., Cell Reports 2022, GEO GSE189033; Miyoshi et al., Nature Genetics 2024, GEO GSE233208. Counts shown are the analysis metadata summarized in this repository.

Full-data local-path analysis resolves cell-type events

3,537
nominal cell-type tests

1,111
nominal condition tests

2,222
cell-type BH calls

0
condition BH calls

- 11,315 eligible cell-type-pair tests and 14,833 condition-within-cell-type tests
- Cell-type calls span 585 blocks and 440 genes
- Every one of 32 null families has zero BH calls for both contrasts
- Pooled null rejection is 0.0500, 0.0100, and approximately 0.0010

Interpretation

The condition family has nominal enrichment but no calibrated BH event. Calibration therefore does not mechanically create a discovery set.

Direct local-path Tealeaf exceeds junction comparators

	Tealeaf			Comparator		
Comparator	Rep.	held	ρ	Rep.	held	ρ
LeafCutter	**50**	**0.799**	**0.711**	26	0.750	0.542
MAJIQ Heterogen	**103**	**0.810**	**0.622**	38	0.698	0.437
scQuint	**101**	**0.819**	**0.656**	56	0.775	0.458

What the paired local-path test changes

Subject-level path ILRs are estimated directly from both primer-specific EC vectors, stabilized by add-two smoothing, variance-moderated across scalar tests, and empirically calibrated with 32 sign-flip families. “Rep.” applies BH to the conjunction p-value on each pairwise shared gene universe; ρ is fold correlation of gene–pair $-\log_{10} p$.

Interpretation

Every aggregated replicated-gene null family has zero BH calls. Add-two smoothing was selected on these splits, so independent-data confirmation remains necessary.

Direct local-path fitting removes the GLMM bottleneck

2.52 h
both subject-half fits

1.03 h
full cell-type fit

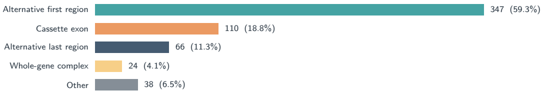
0.288 GiB
peak split worker memory

49×
less worker time
than former full GLMM

- Raw EC likelihood and both primer maps are retained
- No latent mouse effect is optimized separately for every hypothesis
- Each mouse-level composition is fitted once per block and level
- Condition permutations reuse the label-invariant null REML variance

Worker time starts from prepared inputs and excludes shared read processing. The 49-fold comparison is 51.0 versus 1.03 mouse worker hours.

Production cell-type blocks include 110 cassette exons



Functional precedents among production calls

- *Ntrk2*, alternative last region separating short kinase-deficient and long TrkB paths; TrkB.T1 has a characterized astrocyte role
- *Grin1*, C1 cassette exon linked to receptor surface trafficking
- *App*, exon 15 cassette linked to polarized secretion and amyloid processing
- *Dlg4*, alternative-first-region block overlapping functional PSD-95 alpha/beta starts

Boundary of the annotation

A terminal-region label means supported paths differ between a constitutive anchor and a transcript boundary. It does not by itself distinguish promoter choice, terminal exon choice, or polyadenylation.

Takeaways

Model the observations that were measured. Local path effects act through full isoforms, primer-specific EC maps, and observed pseudobulk counts without hard EC allocation.

Calibrate the complete decision rule. Design-matched null families pass through variance moderation, empirical tail mapping, aggregation, and BH.

Local inference realizes the power advantage. The calibrated paired local-path test has more replicated genes and higher held-half replication than every comparator.

Local fitting changes the compute regime. The full cell-type analysis uses 1.03 aggregate worker hours while retaining both primer-specific EC likelihoods.